

Statistical modeling of acute and chronic pain patient-reported outcomes obtained from ecological momentary assessment

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Abstract

Ecological momentary assessment (EMA) allows for the collection of participant-reported outcomes (PROs), including pain, in the normal environment at high resolution and with reduced recall bias. Ecological momentary assessment is an important component in studies of pain, providing detailed information about the frequency, intensity, and degree of interference of individuals' pain. However, there is no universally agreed on standard for summarizing pain measures from repeated PRO assessment using EMA into a single, clinically meaningful measure of pain. Here, we quantify the accuracy of summaries (eg, mean and median) of pain outcomes obtained from EMA and the effect of thresholding these summaries to obtain binary clinical end points of chronic pain status (yes/no). Data applications and simulations indicate that binarizing empirical estimators (eg, sample mean, random intercept linear mixed model) can perform well. However, linear mixed-effect modeling estimators that account for the nonlinear relationship between average and variability of pain scores perform better for quantifying the true average pain and reduce estimation error by up to 50%, with larger improvements for individuals with more variable pain scores. We also show that binarizing pain scores (eg, <3 and ≥ 3) can lead to a substantial loss of statistical power (40%–50%). Thus, when examining pain outcomes using EMA, the use of linear mixed models using the entire scale (0–10) is superior to splitting the outcomes into 2 groups (<3 and ≥ 3) providing greater statistical power and sensitivity.

Keywords: Ecological momentary assessment, Chronic pain, Patient reported outcomes, Statistical modeling, Missing data, Continuous outcomes

1. Introduction

The presence, intensity, duration, and interference with daily activities of painful conditions are a key component of quality of life.⁴⁴ Chronic pain, defined as pain which persists beyond the expected duration associated with an acute injury (eg, 3–6 months) or which co-occurs with a separate chronic condition, is

additionally associated with increased morbidity^{4,9,19} and mortality,^{38,50} although there is substantial heterogeneity in these associations by chronic pain condition. Developing an understanding of the etiology and/or risk factors for developing chronic pain requires researchers to accurately measure pain and use appropriate statistical tools. Here, we discuss key features of participant-reported pain scores that are collected multiple times per participant through ecological momentary assessment (EMA).^{39,55} Pain intensity scores are a key end point for most pain studies, including the Acute to Chronic Pain Acute to Chronic Pain Signatures (A2CPS)⁸ program, the study motivating the current work.

An important outcome for both clinicians and patients is the presence of chronic pain. A clinical diagnosis of chronic pain is often based on several factors including duration and some measure of pain intensity (eg, >3 months), degree of injury (eg, pain out of proportion to the injury), and in some cases pain intensity (ie, >3 for postoperative pain). No universal agreement exists for dichotomizing chronic pain using EMA-assessed pain. Studies analyze EMA assessments of pain in varied ways. Some studies estimate within-individual trends and dynamics of assessment-to-assessment pain, whereas others estimate between-individual differences in average pain. Several approaches for summarizing EMA-recorded pain have been proposed in the literature, ranging from simple summaries such as average pain score^{3,7,10,11,15,20,23,27,28,36,42,47,54} or distributional summaries^{1,24} to more complex approaches such as latent variable/class modelling.^{6,49} Other studies do not calculate

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aggregate measures and instead analyze associations between individual pain measures and other outcome measures.^{2,12,13,17,21,25,28,29,32–36,40,41,43,45,45,48,56} Several studies combined approaches, using both individual pain scores and aggregate scores (eg, average of repeated pain scores in a single day)^{21,46,52} in regression modelling.

For pain scores collected using EMA, there is substantial variability both within and between individuals. Critically, there are substantial differences in the variability of daily pain scores as a function of average pain (averaged across days). For example, participants who have low average pain scores generally exhibit less variability in their pain scores than

participants with higher average pain scores, although at the extremes (very low/high average pain scores) variability is typically low (**Fig. 1**). This violates a key assumption of traditional regression models (constant variance). Failure to account for these data patterns may affect the validity of statistical analyses.

Here, we propose a novel linear mixed model which accounts for the relationship between the variability and average of each individual's pain scores. The proposed model is applied to 2 data sets and used in a simulation study. The simulations quantify the accuracy of several summaries (eg, mean) of pain measures and the effect of thresholding these

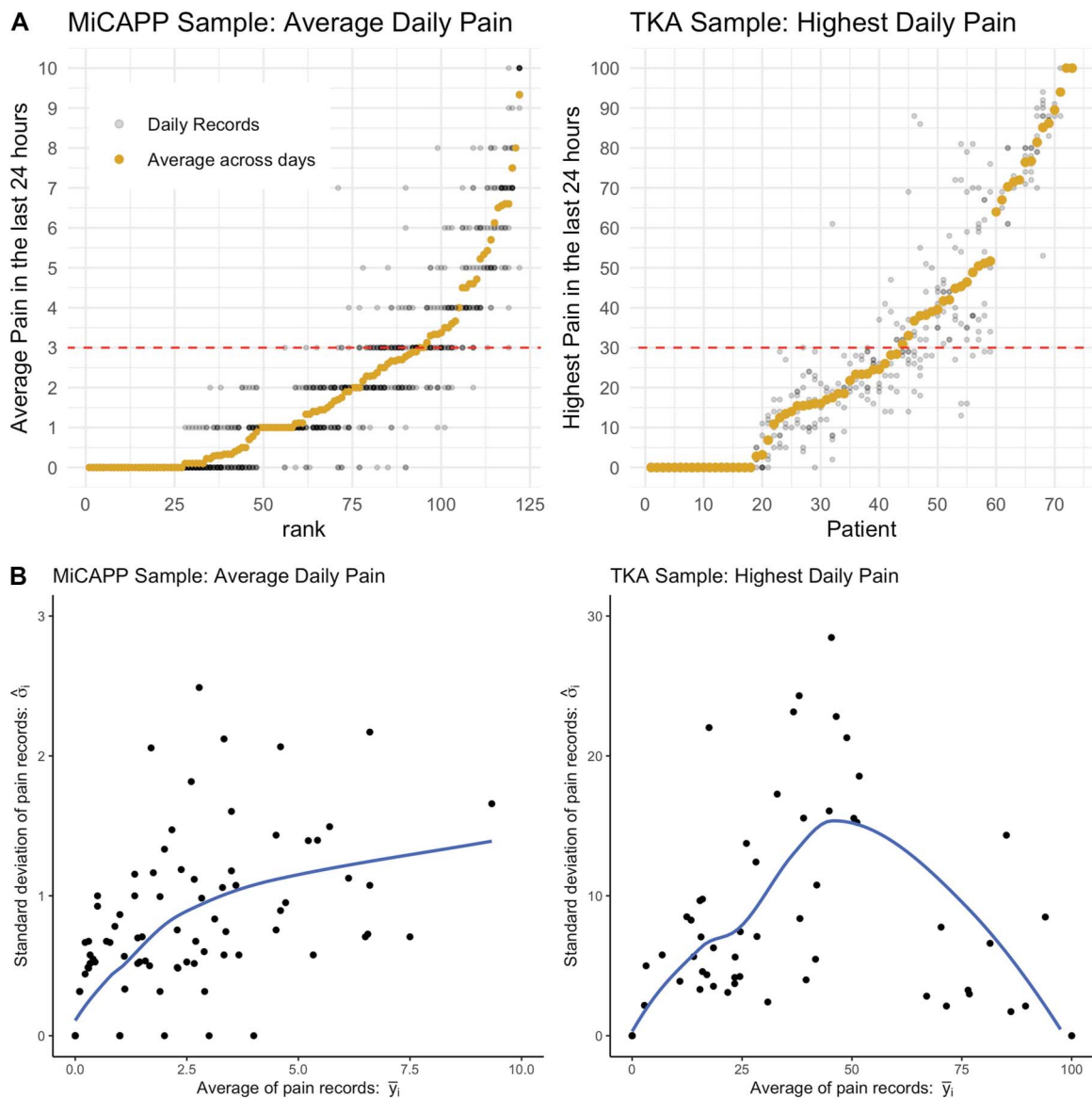


Figure 1. Distribution of daily and average (across days) PROs of average and highest pain in the past 24 hours in 2 clinical samples. (A) Distribution of patients' average and daily pain records and (B) relationship between patients' average pain records and the standard deviation across days. (A) Distribution of patient average daily pain scores (daily records averaged across days of assessment solid gold points) along with daily PRO pain (semitransparent black points). Each column of points on the x-axis represents a single participant, the y-axis corresponds to the participant's average pain records across days (solid gold points) and daily records (semitransparent black points). Participants' data are ordered by their average pain across days. Patient pain is presented for the MiCAPP (left panel) and TKA (right panel) samples which assessed average and highest daily pain, respectively. (B) Patient average pain scores (x-axis) vs standard deviation of pain scores (y-axis) are presented as solid black points. A nonparametric smoother is used to estimate the conditional mean of the standard deviation given average pain (solid blue line). MiCAPP, Mechanisms of the CentrAlized Pain Phenotype; PRO, participant-reported outcomes; TKA, total knee arthroscopy.

summaries to obtain binary clinical end points of chronic pain status (yes/no).

2. Methods

We discuss the quantities of interest that can be derived from EMA assessed pain, followed by approaches for estimating these quantities. First, we introduce some notation that we will use throughout. Mathematical details are kept to a minimum, though a detailed vignette with R code for estimation based on our simulation study presented in Section 4 has been provided in the supplemental material, available at <http://links.lww.com/PAIN/C25>. We denote participants as $i = 1, \dots, N$, each of whom are sampled at $j = 1, \dots, J$ time points (eg, days). Denote the pain score of the i^{th} participant at the j^{th} time point as y_{ij} . Individuals may have missing data, so not all participants will have all J observations. For example, in A2CPS study, we plan to ask $N = 2800$ participants to report their worst pain using the NRS scale (0-10) during the 7 days around 6 months after their surgery.

2.1. Quantities of interest

There are multiple summaries of pain scores (y_{ij}) that may be of interest either scientifically or clinically.⁵³ Most commonly, investigators focus on summaries relating to central tendency, such as the mean (average) or median. One clinical outcome of interest, particularly in the A2CPS study, is whether an individual should be classified as having “chronic pain” as the goal is to develop biomarkers for the development of chronic pain. A definition of chronic pain based on self-reported pain scores, rather than a clinical diagnosis, is challenging for several reasons. One approach is to apply a threshold to a summary of the central tendency of participants’ pain scores. For example, chronic pain status may be defined as observing an average pain score of at least 3 on a 0 to 10 scale. However, there is greater uncertainty in classification of individuals who have highly variable pain scores or whose average is close to the threshold. As a result, it may be of interest to quantify uncertainty in the classification itself (uncertainty in the estimated participant average pain scores).

The simplest estimators use only a participants’ own data for estimation. However, it may be possible to borrow information across participants by fitting a statistical model to the data. In addition, for many models, these quantities can be assigned a measure of uncertainty. For example, we could construct 95% confidence intervals for a participants’ average pain scores given the observed data and estimated model parameters, where parameters are estimated using all participants’ data. Although it is possible to construct 95% confidence intervals for the mean and the median using only participants’ own data, when the number of subject-specific measurements (J) is small, the resulting confidence intervals are often highly variable or rely on asymptotic results, resulting in unnecessarily wide confidence intervals or intervals which do not have nominal coverage (ie, more or less than 95%).

Below, we discuss summaries of potential clinical and scientific interest which can be broadly classified into 3 categories: central tendency, extreme value, or threshold-based summaries.

2.1.1. Central tendency

The most used measures of central tendency are the mean (expected value) and the median (50th percentile). The simplest estimators of the mean and the median are the corresponding empirical estimators, specifically the subject-specific sample

average ($\bar{y}_i$) and the sample median ($\hat{\tau}_i^{50}$). Model-based estimators of these quantities may also be obtained using, eg, linear mixed models (see Section 2.2).

2.1.2. Extreme values (maximum)

It may also be of interest whether a participant reports any pain scores above a particular threshold, potentially representing the presence of extreme pain episodes. For example, some pain conditions, such as migraine, involve intermittent, but intense, pain episodes. In such cases, researchers may be interested in whether an individual reports a pain score greater than 8 for any day or the numeric value of the maximum pain score. The distribution of the maximum observed pain score for an individual i depends on the amount of missing data present. Alternatively, one can estimate the probability of observing a maximum pain score above a certain threshold for a fixed number of EMA pain records. Subject-specific estimands of this type are estimable, but may be highly variable. By contrast, model-based estimators of these quantities are available, but rely on distributional assumptions (eg, independence within subjects, constant variance, etc.) for accuracy.

2.1.3. Threshold-based summaries

In each of the above examples (central tendency, extreme values), it may be of interest whether the estimate for an individual is above or below a specific clinically meaningful threshold. For example, it has been proposed to use a threshold of an average pain score of >3 of 10 to assess an individual’s chronic pain status. More generally, any feature/statistic (eg, sample median) can be thresholded. Individuals with values of the statistic near the threshold or individuals with more variable pain records that cross the threshold value will tend to have greater uncertainty in their estimate because of increased variability in the observed statistic, thus placing them at greater risk for misclassification.

2.2. Statistical models

The accuracy of both point estimates and inference based on statistical models depends on how appropriate assumptions are. We investigate the performance of 2 linear mixed-effects models in estimating patient-specific quantities of interest. Alternative models may be more appropriate for different patient populations. For example, one might consider modelling these data using an ordinal mixed model approach,²⁵ though such approaches are computationally intensive and require nontrivial statistical, programming, and mathematical expertise.

2.2.1. Constant variance linear mixed model

The first model we propose is the classical Gaussian random intercept model, where both the subject-specific deviations (b_i) from the global average (β_0) and the daily deviations from subject-specific averages (ϵ_{ij}) are normally distributed with variances σ_b^2 and σ_ϵ^2 , respectively. This model assumes that the variance is constant within subjects (ie, all participants’ pain scores have the same variability around their mean). Thus, we refer to this as the “constant variance linear mixed model” (LMM-CV). The model is expressed as follows:

$$y_{ij} = \beta_0 + b_i + \epsilon_{ij}$$

$$b_i \sim N(0, \sigma_b^2)$$

$$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

The LMM-CV model is estimated using maximum likelihood approaches implemented in the *lme4* package⁵ in *R*.

2.2.2. Nonconstant variance linear mixed model

The LMM-CV model described in Section 2.2.1 does not appropriately model the mean variance relationship observed in EMA-assessed pain data (see Section 3). Specifically, there is a nonlinear relationship between the mean and variance induced by the bounded nature of the data, with individuals reporting average pain scores close to the boundaries (eg, 0 or 10 on a 0–10 scale) having lower variance in their responses than individuals away from the boundary. To account for this phenomenon, we propose to model the mean–variance relationship explicitly, which we call the “nonconstant variance linear mixed model” (LMM-NCV). Specifically, we model the data as follows:

$$y_{ij} = \beta_0 + b_i + \epsilon_{ij}$$

$$b_i \sim N(0, \sigma_b^2)$$

$$\epsilon_{ij} \sim N(0, \sigma_{y,i}^2)$$

$$\log(c + \sigma_{y,i}) | b_i, \beta_0 \sim N(f(\beta_0 + b_i), \sigma_\epsilon^2)$$

The log transformation of the subject-specific standard deviation corresponds to an assumption that subject-specific standard deviations follow a log-normal distribution, with c being a small constant which prevents poor boundary behavior for $\sigma_{y,i}^2$ near 0. The relationship between subject-specific mean and variability, $f(\cdot)$, may be modelled parametrically (eg, using a polynomial function or linear splines) or nonparametrically (eg, using penalized splines). This model allows for heterogeneity in variability within individuals who have the same mean, where the degree of heterogeneity is increased when $f(\cdot)$ is larger (eg, individuals with average pain scores between 3 and 5).

We estimate the LMM-NCV using a Bayesian approach. Briefly, Bayesian statistics requires a prior distribution on all model parameters (σ_ϵ^2 , σ_b^2 , β_0 , and the parameters associated with $f(\cdot)$). Given the assumed prior distribution on model parameters and the observed data, inference may be made on parameters by the posterior distribution of all model parameters given the data.

For the purposes of this manuscript, we consider only the parametric linear spline model

$$f(\beta_0 + b_i) = \alpha_0 + \alpha_1 \mu_i + \sum_k \alpha_k (\mu_i - \kappa_k)$$

where $\mu_i = \beta_0 + b_i$ and κ_k are the knot locations.

For the linear spline basis, however, the method readily extends to alternative parametric and nonparametric forms.^{16,31} The choice of a linear spline basis was made because (1) the linear spline basis works well for our data and (2) nonparametric variations of the LMM-NCV model are quite challenging given our data.

Before model estimation we rescale all pain scores to the [0, 1] scale. For our data applications and simulations, we found that the knot locations $\kappa_1 = 0.2$, $\kappa_2 = 0.5$ provide the needed flexibility for modelling the nonlinear mean–variance relationship present in our data. In addition, we use weakly informative normal prior distributions for β_0 , α_0 , α_1 , and α_2 and weakly informative half-normal prior distributions for σ_ϵ and $\sigma_{y,i}$. We provide code for estimating the proposed LMM-NCV for fitting this model using the STAN programming language¹⁴ using in the *rstan* package in *R*.

2.3. Data description

We apply our method to 2 different clinical samples which collected EMA pain measurements on participants. They are described in detail in Sections 2.4.1 and 2.4.2 below. We performed exploratory analyses to evaluate the mean–variance relationship in the 2 samples by plotting the patient-specific average and standard deviation of PRO pain scores across days. Data were examined for systematic patterns of missing data. Within each data set, empirical and model-based estimates of measures of central tendency (mean/median) were calculated.

Pair-wise cross tabulations of classifications of chronic pain across estimators (confusion matrices) were constructed to assess rates of differential classification for threshold-based measures of central tendency.

2.3.1. Total knee arthroscopy data

Our first data application involved participants who underwent total knee arthroscopy (TKA). The study collected the PROs asking about the highest and average daily pain and percent of the day in pain and interference of pain with work or other activities. In our primary analysis, we will focus on highest daily pain because this is the outcome of the A2CPS program, though the supplemental material (available at <http://links.lww.com/PAIN/C25>) contains results for other PROs. Pain was assessed on a numeric rating scale between 0 and 100 for 7 consecutive days using EMA 6 months postsurgery. Participants completed surveys in the evening and provided ratings (ie, pain, intensity) for the preceding day. The current sample contains data from $N = 73$ participants with a total of 411 days of complete data. For creating a threshold-based measure of chronic pain, a threshold of $c = 30$ was used.

2.3.2. Mechanisms of the CentrAlized Pain Phenotype data

The Mechanisms of the CentrAlized Pain Phenotype (MiCAPP) study collected information on the average daily pain over 10 days using EMA in participants with chronic musculoskeletal pain. Pain was assessed on a scale between 0 and 10. Each patient received the 10-day pain and sleep questions via text or email starting 10 days prior to their scheduled surgery date and then again 10 days before the 3-month time point. Patients in the imaging cohort received the postoperative EMAs 10 days before the scheduled 3-month study visit instead of the 3-month time point. These questions were prescheduled in Qualtrics to send once per day at 3:00 PM and were set to expire 24 hours later. If a patient was not completing these daily surveys, the research team would attempt to reach out by phone to remind the patient of the surveys or complete it with them over the phone. The current sample contains data from $N = 122$ participants with a total of 943 days of complete data from the 3-month postoperative assessments. For creating a threshold-based measure of chronic pain, a threshold of $c = 3$ was used.

2.4. Simulation study

In real data with no established “gold standard” for classifying chronic pain, we cannot assess the relative accuracy of various classifications or estimators of subject-specific distributional quantities. Moreover, the potential impact of binarization on downstream analyses cannot definitively be quantified. Thus, we conducted a simulation study to investigate these issues in greater detail.

The goals of the simulation study are to (1) compare the relative performance of empirical and model-based (LMM-CV, LMM-NCV) estimators of central tendency and inference (model based estimators only); (2) evaluate the classification accuracy of binarizing individuals as having chronic pain using a threshold based on estimated measures of central tendency; (3) quantify the information loss associated with binarizing the continuous summaries of central tendency using various estimators in the context of regression modelling; and (4) assess the sensitivity and relative performance of various estimators for goals (1)-(3) under various missing data generating mechanisms. Below, we describe the data-generating mechanism and evaluation criteria for the simulation study.

2.4.1. Data-generating mechanism

2.4.1.1. Pain scores

The simulation study uses parameters of the LMM-NCV model (σ_e , σ_b , β_0 , α) estimated from the TKA sample to generate simulated data similar to the real data. Specifically, we fit the LMM-NCV model to the TKA data and used the posterior means for each parameter as the true parameters in the data generating mechanism. Pain scores are simulated on the 0 to 1 scale used for estimation but are presented in Section 3 on the 0 to 10 scale for comparability with the data application results.

For each simulation scenario, we assessed performance using 2 sample sizes, $N = 100$ and $N = 200$, with a maximum of $J = 10$ and $J = 20$ observations per individual, respectively. In addition, we simulated data under 2 missing data-generating mechanisms: (1) noninformative missingness (ie, missing data not associated with pain score) and (2) informative missingness (ie, missing data associated with pain score). The current simulation study only considers missing data mechanisms which are related to individuals' average pain scores, rather than daily pain scores. Based on these choices (sample size, number of observations, and missingness mechanism), we generated data under 8 different simulation scenarios.

2.4.1.2. Missing data

For noninformative missing data scenario, data were simulated such that every observation for every individual has a 60% probability of being missing. For the informative missingness scenario, we generate missing with a higher probability of missing data near the threshold for chronic pain (highest for individuals with average pain score of 0.3 in the scale of the simulated data, 3 on a 0-10 scale) and decreasing probability of missingness toward the upper and lower end of the range.

2.4.1.3. Association with a biomarker

In addition to simulating pain scores, we simulate a single "biomarker" for each participant (X_i) which is associated with the individuals' average pain scores. We simulate biomarkers which are associated with either the continuous subject average (μ_i) or the thresholded subject average pain scores ($1(\mu_i \geq c)$, with $c = 3$ on a 0 to 10 scale). In both cases, the biomarkers are simulated given the pain score data generated as described above. The biomarker simulation is intended to provide a simplified example of how binarization and statistical modelling affects the ability to detect associations with EMA-assessed pain, a primary goal of the A2CPS initiative, using a single continuous biomarker. In the A2CPS study, many biomarkers of mixed data types derived from multiple high dimensional data sources (imaging, omics, etc.) are

candidate biomarkers, but a simpler simulation study provides insights into relevant mechanisms and issues that may arise when studying more complicated data sources.

For continuous subject average models, we simulate data as $X_i = (\mu_i - \epsilon_i)$ which corresponds to the linear regression model $\mu_i = \beta_0 + \beta_1 X_i + \epsilon_i$ where $\beta_0 = 0$, $\beta_1 = 1$, and $\epsilon_i \sim N(0, \sigma_\epsilon^2 = 1)$. For the threshold subject average pain scores, we simulate data using the principles of linear discriminant analysis. Specifically, we simulate $X_i | 1(\mu_i \geq c) \sim N(\mu_X, \sigma_X^2 = 1)$ with $\mu_X = -0.3$ if $\mu_i < c$ and $\mu_X = 0.3$ if $\mu_i \geq c$. This corresponds to the logistic regression model $\text{logit}\{\Pr(\mu_i \geq c)\} = \beta_0 + \beta_1 X_i$, where $\beta_0 = 0$ and $\beta_1 = 0.6$.

For each data-generating models, associations are estimated using the continuous or binarized estimates from each of the empirical and model-based estimators of μ_i using linear and logistic regression models, respectively. That is, for each true model, we fit both a correctly specified model and an incorrectly specified model, using the predicted pain outcomes μ_i , $1(\mu_i \geq 0.3)$ for each estimator and each simulated data set. This allows for comparing the effect of binarizing the predicted subject mean pain scores even when the true association is continuous.

2.4.2. Evaluation criteria

2.4.2.1. Estimation and inference on pain scores

For each scenario, we simulated 200 data sets. For each simulated data set, we assessed the relative performance of each estimator using mean squared error (MSE), defined as $N^{-1} \sum_{i=1}^N (\hat{\mu}_i - \mu_i)^2$ (all estimators), 95% coverage probability (model-based estimators), and threshold inference on subject observations $\Pr(y_{ij} \geq 0.3)$ (model-based estimators). Each of these criteria were evaluated globally and locally as a function of μ_i to assess whether certain models performed better for certain true participant average pain score. Note that $\Pr(y_{ij} \geq 0.3)$ may be used to calculate summaries related to extreme value measures (eg, the probability the maximum of 7 pain scores is above 8).

2.4.2.2. Classification

In addition to the estimation and inference accuracy discussed above, we use estimates for μ_i to classify individuals as either "chronic pain" or "no chronic pain" based on whether $\mu_i \geq 0.3$. We evaluate the estimators' classification performance using accuracy (% correctly classified), sensitivity, specificity, negative predictive value (NPV), and positive predictive value (PPV). As with the estimation and inference criteria, these measures are calculated both globally and locally.

2.4.2.3. Association with a biomarker

We assess the impact of the choice of estimator and binarization on the association between pain scores and a biomarker using statistical power (probability of rejecting the null hypothesis of no association when there is an association) in each of the models discussed in Section 4.1. Statistical power is assessed using the two-sided Wald tests on regression coefficients at a type-I error rate (α) of 0.05.

3. Results

3.1. Data application

The observed data from the MiCAPP and TKA samples are presented in **Figure 1** and show the variability in pain scores both within and between subjects, termed the mean-variance

Table 1

Confusion matrix for classification of chronic pain.

	Sample median		LMM-NCV		LMM-CV	
	Yes	No	Yes	No	Yes	No
TKA sample						
Sample mean						
Yes	42	1	43	0	43	0
No	3	27	0	30	0	30
MiCAPP sample						
Sample mean						
Yes	95	0	94	1	95	0
No	5	22	0	27	0	27

Chronic pain is determined using a threshold of >30 and >3 in the TKA and MiCAPP samples, respectively. Confusion matrices are presented for the sample median (columns 1-2), the nonconstant variance linear mixed model (LMM-NCV, columns 3-4), and the constant variance linear mixed model (LMM-CV, columns 5-6) in comparison with the sample mean (rows 1-2). MiCAPP, Mechanisms of the CentrAlized Pain Phenotype; LMM-CV, constant variance linear mixed model; LMM-NCV, nonconstant variance linear mixed model; TKA, total knee arthroscopy.

relationship. **Figure 1A** shows individual pain scores (semi-transparent black points) along with average values (solid gold points) ordered by participant average pain score (x-axis, increasing from left to right) in the MiCAPP (left panel) and TKA (right panel) samples. **Figure 1B** presents the average pain score (x-axis) vs the standard deviation (y-axis) of pain scores across days for the MiCAPP (left panel) and TKA (right panel) samples. **Figure 1A** illustrates that there are a number of participants who consistently report 0 pain in both the MiCAPP ($\approx 22\%$) and TKA ($\approx 25\%$) samples. In addition, there are a lower proportion of

individuals whose average is near the proposed thresholds of 3 and 30 in the MiCAPP and TKA samples, respectively, used to define the presence of chronic pain with a binary scale. Specifically, 16% and 7% of patients have average pain scores within ± 0.5 and ± 5 in the MiCAPP and TKA samples, respectively. However, we see that daily pain scores tend to be variable at or near the proposed thresholds for classifying chronic pain in the 2 samples. **Figure 1B** provides a more direct illustration of the mean–variance relationship inherent in these data. Specifically, there is a nonconstant and nonlinear relationship between patient average pain and day-to-day variability. This nonlinearity is most pronounced in the TKA sample, with a clear inverse U-shaped association whereby variability peaks for participants with average pain scores near the midpoint.⁵⁰ In the MiCAPP sample, there is only one individual with average pain score above 8, and this person reported a pain score of 10 every day except one, where they reported an average daily pain of 5. Thus, it is difficult to assess the behavior in participants with high average pain scores in this cohort. Figure 5 presents a plot similar to Figure 1 for pain ratings other than highest daily pain (average daily pain, percent of the day in pain, and pain interference). The findings confirm the inverse U-shaped association between mean and variance exists for other pain measures collected in the TKA sample. **Table 1** presents the confusion matrices associated with classifying participants as having chronic pain at a threshold of 3 and 30 in the TKA (top) and MiCAPP (bottom) samples, respectively, using different estimators of central tendency as compared with the sample average. In these 2 samples, the sample mean has a nearly

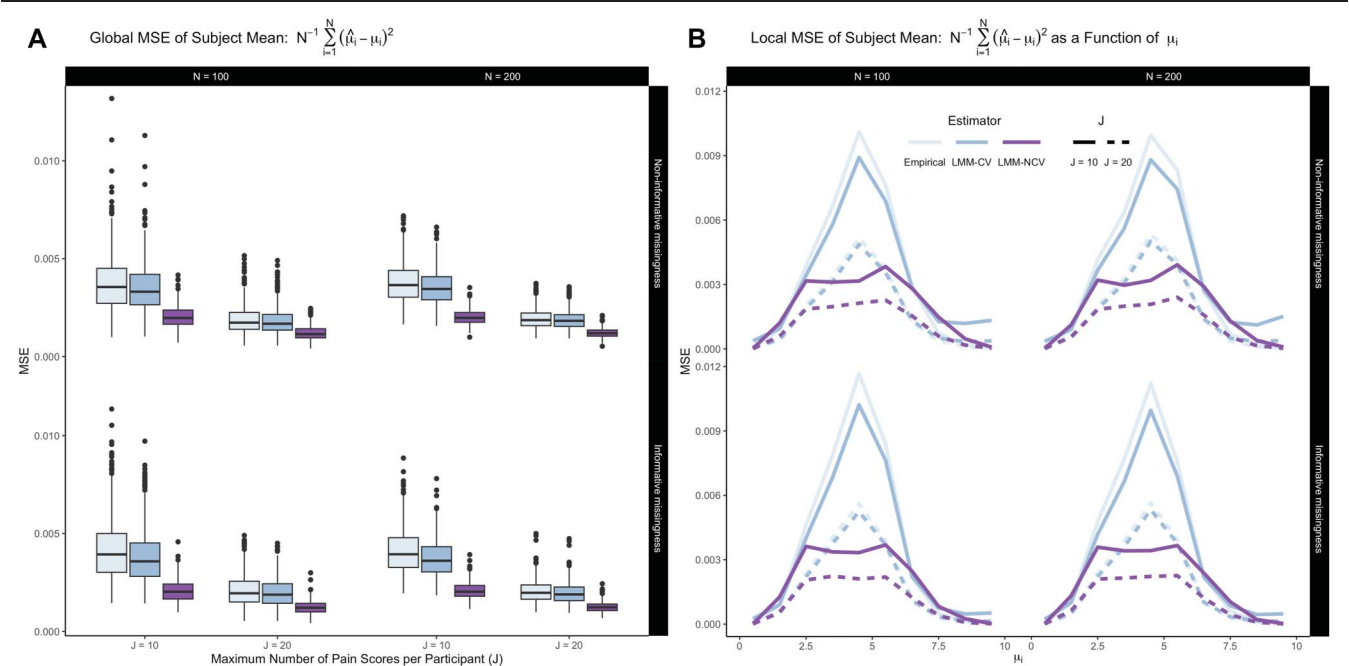


Figure 2. Estimation accuracy of various estimators for μ_i , participant specific average pain score as measured by mean squared error (MSE). Color corresponds to a particular estimator: empirical, LMM-CV, and LMM-NCV are presented as light blue, blue, and purple, respectively. Within (A–C), each panel corresponds to a different number of participants in the data set (columns, $N = 100$ or $N = 200$, left to right) and a different missing data generating mechanism (rows, noninformative or informative, top to bottom). Within (D) each column corresponds to MSE grouped by number of observed pain scores (1–3, 4–7, 8–10, left to right). (A) Boxplots of MSE averaged across all individuals within a simulated data set. The x-axis corresponds to the number of maximum pain scores per person (J) of either 10 or 20. (B) Average MSE across all data sets calculated separately for individuals with true average μ_i binned into 10 intervals of length 0.1. Line type corresponds to J of either 10 (solid lines) or 20 (dashed lines). (C) Boxplots of MSE averaged across all individuals within a simulated data set among scenarios with maximum number of pain scores per person of 10 ($J = 10$). The x-axis corresponds to number of observed pain scores per person (1–3, 4–7, 8–10, left to right). (D) Average MSE across all data sets calculated separately for individuals with true average μ_i binned into 10 intervals of length 0.1 for scenarios with $N = 100$, $J = 10$. LMM-CV, constant variance linear mixed model; LMM-NCV, nonconstant variance linear mixed model.

100% rate of agreement with the 2 linear mixed models: linear mixed model with constant variance (LMM-CV) and linear mixed model with nonconstant variance (LMM-NCV). The sample median agrees nearly as well, with agreement rates of 95% and 96% in the TKA and MiCAPP samples, respectively.

3.2. Simulation study

3.2.1. Estimation and inference on pain scores

Figure 2 presents the estimation accuracy criteria measured by MSE both globally (**Figs. 2A and C**) and locally (**Figs. 2B and D**) as a function of the number of participants (columns, **Figs. 2A–C**), number of observed pain scores per participant (columns), and missing data-generating mechanism (rows). Lower MSE scores represent better accuracy.

In these data, the LMM-NCV (purple) substantially outperforms both a classical linear mixed model assuming constant variance (LMM-CV, blue) and the empirical average (light blue) both globally (**Figs. 2A and C**) and locally (**Figs. 2B and D**). Across methods accuracy improves with increasing both the number of participants (N) and number of observations per participant (J). The degree of improvement is strongest for participants whose individual pain scores are most variable, which include those near the threshold for classifying chronic pain status.³ In addition, estimation accuracy improves with an increasing number of observed scores both globally (**Fig. 2B**) and locally (**Fig. 2D**). These simulations suggest that the LMM-NCV has benefits both for average/median MSE and the reduction of outlying MSE values for all number of observations per person. Furthermore, with at least 4 observations per participant, all methods have reasonable accuracy, but the LMM-NCV outperforms the other methods by nearly 50%.

Figure 3 presents the estimation accuracy for $\Pr(y_{ij} \geq 3)$, the probability an individual observation is above the threshold for chronic pain (**Fig. 3A**), and coverage rates of the 95% confidence/credible intervals for the participant average pain score (**Fig. 3B**) for the LMM-CV (light blue) and LMM-NCV (blue) models. **Figure 3** presents the confidence/credible interval accuracy for scenarios with $N = 100$ participants and maximum number of 10 pain scores by number of observed pain scores per person (columns, 1-3, 4-7, 8-10, left to right) and missing data-generating mechanisms (rows, noninformative and informative, top to bottom). These results show that the LMM-NCV model outperforms the LMM-CV model in estimation accuracy for all number of observations per participant. The degree of outperformance for estimating $\Pr(y_{ij} \geq 3)$ markedly increases with the number of observations per participant. In addition, while near nominal coverage (ie, 95% confidence intervals contain the true parameter value 95% of the time) is achieved for the LMM-NCV model for all values of μ_i when individuals have at least 4 pain scores, the LMM-CV model exhibits sizable under coverage for individuals with true average pain (μ_i) in the middle of the scale (confidence intervals too short for μ_i between 3 and 6) and sizable over coverage for individuals with the true average pain at the ends of the scale (confidence intervals too wide for μ_i 0-3 and 6-10).

3.2.2. Classification

Figure 4 presents the results of classification of chronic pain using the same visual structure as **Figure 2** but only considers the case where study sample size is $N = 100$ (the results for $N = 200$ are similar). The results largely agree with the findings on reduced MSE using the LMM-NCV model, although all methods perform well. Interestingly, the LMM-NCV model performs worse than the LMM-CV model for individuals with average pain scores between

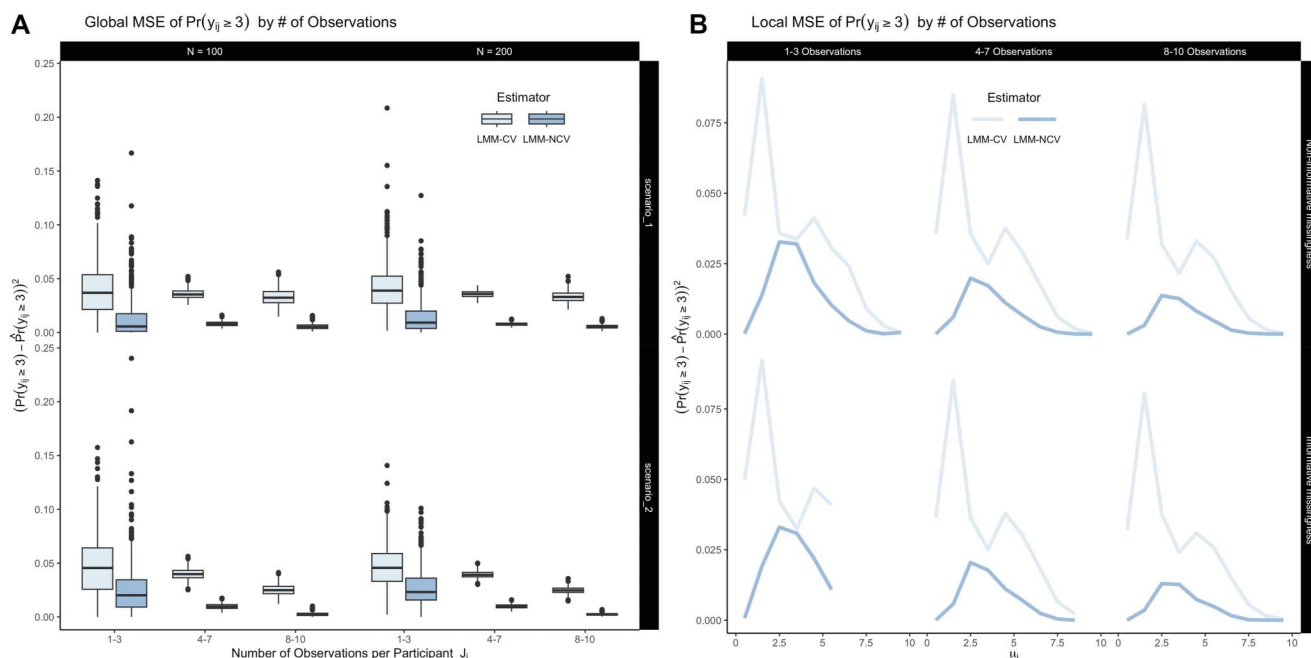
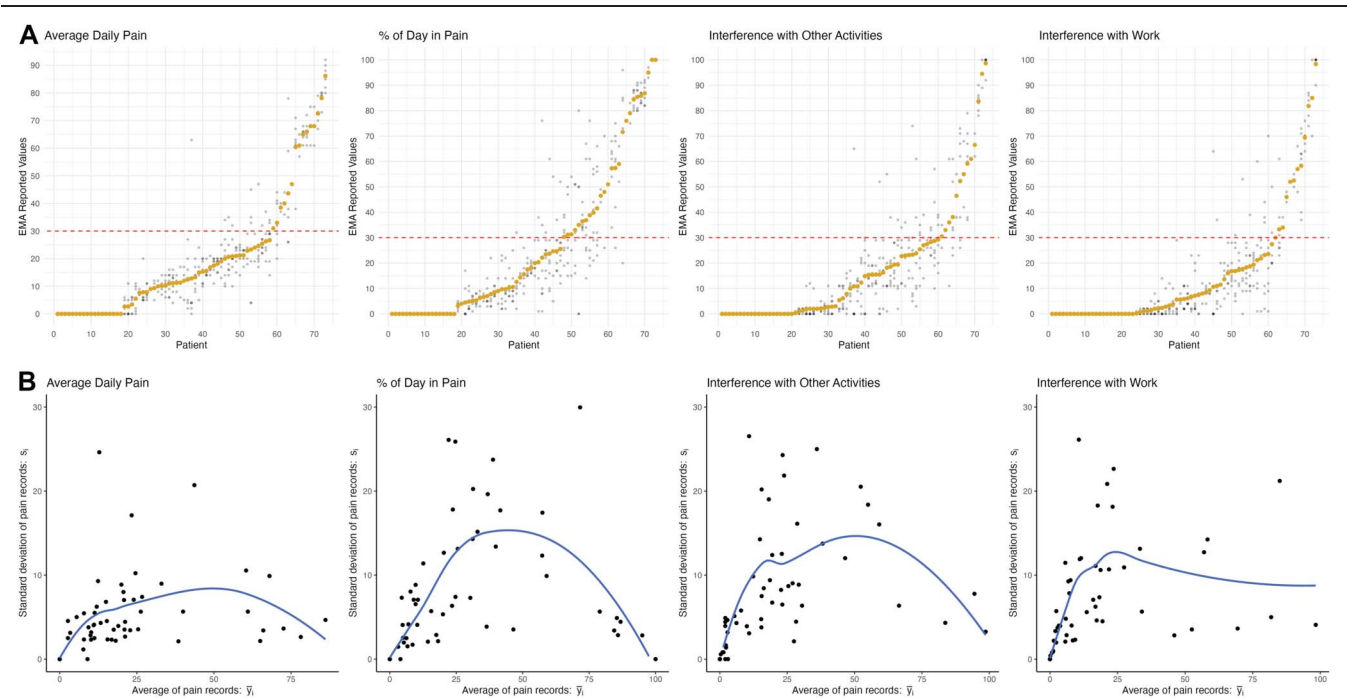
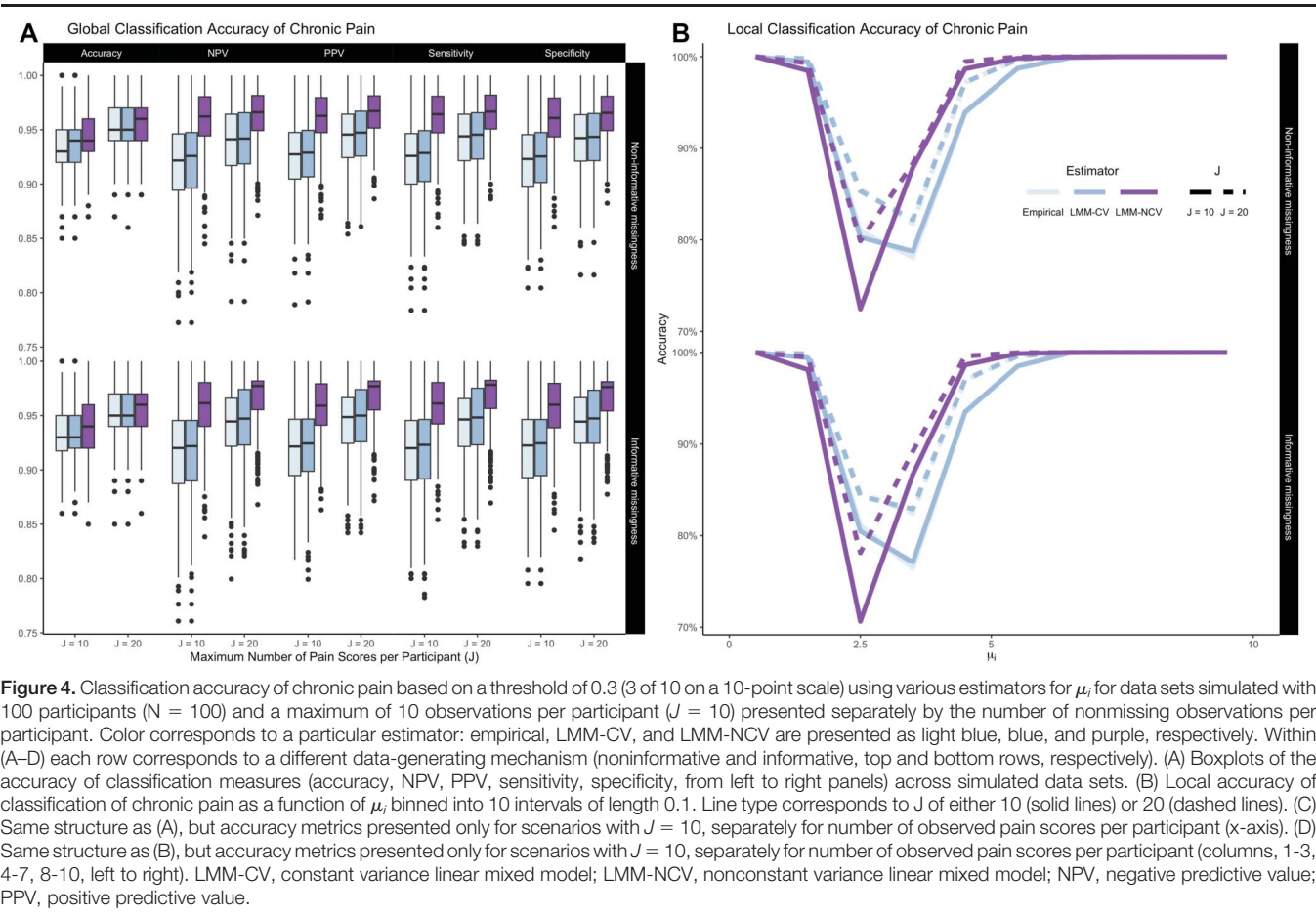


Figure 3. Inferential accuracy of various estimators for (A) $\Pr(y_{ij} \geq 0.3)$ and (B) 95% coverage probabilities for scenarios with maximum number of pain scores equal to 10 ($J = 10$) and number of persons equal to 100 ($N = 100$). Accuracy is presented as a function of individuals' true average pain score (x-axis, μ_i). Color corresponds to a particular estimator: LMM-CV and LMM-NCV are presented as light blue and blue, respectively. Within (A and B), each panel corresponds to a different number of observed pain scores per person (columns, 1-3, 4-7, 8-10, left to right) and a different missing data-generating mechanism (rows, noninformative or informative, top to bottom) (A) MSE averaged across all simulated data sets. (B) Coverage probability averaged across all simulated data sets. Quantities were calculated by binning individuals into 10 intervals of length 0.1 individuals within a simulated data set. LMM-CV, constant variance linear mixed model; LMM-NCV, nonconstant variance linear mixed model; MSE, mean squared error.



2 and 3. This can be attributed to the fact that the LMM-CV model tends to shrink individuals' estimates more in the region with highest variability, approximately 4 to 5. That is, in the LMM-NCV model, individuals with average pain scores slightly below 3 have the estimator shrunk slightly above the threshold of 3. However, **Figure 2** indicates that μ_i is better estimated by the LMM-NCV model (**Fig. 5**).

3.2.3. Association with a biomarker

Table 2 presents the estimated power to detect the association between the biomarker and individuals' pain scores. The results are organized by missing data-generating mechanism, true data-generating mechanism (continuous or binarized), estimated model (continuous or binarized), number of subjects in each simulated data set ($N = 100, 200$), and maximum number of pain scores per subject ($J = 10, 20$). Within each of these nested groups, study power is presented for each estimator separately by row (Empirical, LMM-CV, LMM-NCV, from top to bottom). Within a given simulation scenario and model fit, LMM-NCV tends to provide slightly higher power than the sample average (empirical) or LMM-CV, although this is not uniformly true. More importantly, regardless of the true data-generating mechanism, there is a substantial loss of power (approximately 40%-50%) across the scenarios considered here. That is, for the purposes of statistical power in these simulations, the choice of estimator is not as impactful as the decision to binarize the estimated average pain scores.

Table 2
Simulation study findings on statistical power (type II error probability) to detect associations between a biomarker and individuals' pain scores.

No. of patients (N)								
Estimator	100		100		200		200	
Number of observations per patient (J)								
	10	20	10	20	10	20	10	20
Missing data scenario 1: noninformative missingness								
Estimated model: continuous					Estimated model: binary			
True model: continuous								
Empirical	0.658	0.622	0.928	0.936	0.316	0.315	0.515	0.510
LMM-CV	0.654	0.622	0.930	0.936	0.317	0.312	0.514	0.511
LMM-NCV	0.664	0.626	0.936	0.938	0.313	0.310	0.492	0.490
True model: continuous								
Empirical	0.680	0.650	0.926	0.925	0.322	0.314	0.523	0.523
LMM-CV	0.680	0.650	0.926	0.925	0.323	0.315	0.518	0.521
LMM-NCV	0.682	0.656	0.926	0.919	0.322	0.300	0.493	0.501
Missing data scenario 2: informative missingness								
Estimated model: continuous					Estimated model: binary			
True model: continuous								
Empirical	0.839	0.827	0.961	0.966	0.380	0.391	0.499	0.498
LMM-CV	0.84	0.828	0.961	0.962	0.380	0.394	0.502	0.499
LMM-NCV	0.829	0.831	0.957	0.963	0.405	0.401	0.524	0.525
True model: continuous								
Empirical	0.650	0.680	0.905	0.935	0.320	0.320	0.502	0.513
LMM-CV	0.650	0.685	0.910	0.935	0.312	0.323	0.507	0.513
LMM-NCV	0.665	0.680	0.930	0.935	0.290	0.325	0.490	0.482

LMM-CV, constant variance linear mixed model; LMM-NCV, nonconstant variance linear mixed model.

4. Discussion

We have presented a study of several alternative approaches for summarizing EMA-assessed pain and a comparison of various empirical and model-based estimators for quantities of interest, including a novel method which explicitly models the nonlinear mean–variance relationship observed in the data. We provide 2 real data applications and a simulation study which show the relative performance of various estimators of distributional quantities and inference.

Classification of chronic pain using threshold-based summaries (eg, average pain of at least 3 of 10 on a 10-point scale) is an outcome which has appeal as a clinically meaningful and interpretable end point. In simulations, both model-based and empirical estimators of subject-specific central tendency (ie, mean, median, mode) perform well and lead to reasonably low rates of misclassification for threshold-based quantities (eg, labelling individuals as having chronic pain). Misclassification rates of “chronic pain” for all approaches is low for individuals with at least 3 observed pain scores, with approximately 80% of individuals with true average pain near the threshold for classification (3 on a 10-point scale) being correctly classified. This result is consistent with previous findings on the reliability of average pain scores.⁵¹

However, when considering the accuracy of the estimated average pain, model-based estimators which account for the relationship between the average pain score and variability of individual pain scores present in this type of data (eg, the proposed LMM-NCV model) outperform alternative, simpler estimators, including a classical linear mixed model with a random intercept (LMM-CV). The improvement is particularly strong for participants with more variable pain scores, with a reduction in MSE/error of up to 3 times. An interesting finding is that model-based estimators that do not appropriately model the relationship between the average pain score and the variance in the pain scores may poorly estimate the average pain score of individuals near the boundary of the scale (eg, 0 or 10 on a 0-10 scale). Moreover, our simulation study shows that estimates of confidence intervals and probabilities of individual pain scores can be incorrect when not taking the mean–variance relationship into consideration. Specifically, a classical random intercept model (eg, LMM-CV) will tend to underestimate (or overestimate) uncertainty on subject-specific average pain scores for those individuals who have more (or less) variable pain scores. Because individuals near commonly used thresholds for chronic pain tend to have more variable pain scores, this may lead to artificially increased certainty of classification of individuals' threshold quantities for “borderline” individuals. In the context of pain research, this may be relevant for studies where there are a sizable proportion of participants who are near the threshold used for binarizing the data or when participants near the threshold are of primary interest. In these scenarios, uncertainty estimates may be used in subsequent analyses to account for the uncertainty in estimates of subject average pain scores. Inappropriate modelling of pain scores would tend to underestimate variability and potentially result in spurious findings.

Statistical power is substantially reduced when binarizing continuous summaries of pain, even when the true association is between individuals' binarized chronic pain status and the biomarker. In our simulations, this decrease in power is on the order of 40% to 50%. This reduction in power is likely attributable to the increased uncertainty in individual classifications for those individuals near the threshold for classifying chronic pain, which are reflected in their estimated/predicted average pain scores.

This loss of power will increase with misclassification rates, which are more likely to occur in populations with a high prevalence of individuals with average pain near the binarization threshold. Moreover, statistical inference may be negatively affected by the failure to correctly model uncertainty in the binarization procedure. Although such uncertainty can be incorporated in statistical modelling, within-patient variability must be correctly modelled.

There are several limitations of our work. First, although we introduced a new model to account for the mean–variance relationship observed in EMA-assessed pain scores in our 2 data applications, this may not be the optimal approach in all populations. Alternative approaches, such as transformations of EMA-assessed pain scores or generalized linear mixed models, may offer further improvements in subject-specific predictions. However, such models entail increased methodologic and computational complexity, particularly for obtaining subject-specific inference, leading to barriers to their use in practice. In addition, none of the methods proposed here directly account for scenarios where data are missing not at random. Addressing this problem is beyond the scope of the current work, but missing data patterns should be rigorously investigated in clinical applications. Finally, our simulation study shows that choice of estimator is not particularly important for statistical power as long as the data are not binarized, despite the substantially higher accuracy for the LMM-NCV approach. This is likely due to the extremely high correlation between true subject-specific average pain scores and predicted values from all continuous estimators. It may be that incorporating subject-specific variability estimates in regression modelling (eg, weighted least squares, hierarchical Bayesian models for both pain scores and biomarker associations) may improve the power of the LMM-NCV approach. Moreover, variability in pain scores may be informative, and methods which estimate both subject-specific mean and variability may play a key role in identifying “clusters” of patients with common pain experiences.

5. Conclusion

In summary, our results show that subject-specific empirical estimators (ie, subject average pain score) are likely sufficient for investigators whose interest lies in using point estimates for central tendency or classification of chronic pain based on these estimates. However, model-based estimators, like the LMM-NCV presented in this manuscript, which model the mean–variance relationship in EMA-assessed pain scores offer the potential for nontrivial increases in estimation and inferential accuracy. Although obtaining a higher number of observations per participant improves results, collecting 4 observations per individual provides large enough data points for all methods. We recommend against binarizing the data. Rather, we recommend performing statistical analyses on the continuous data because of the dramatic reduction in statistical power associated with binarization. We sum up the findings and implications of our work for researchers in 4 key points: (1) if the goal of an analysis is simply to obtain point estimates for pain, sophisticated statistical modelling of the data is likely unnecessary; (2) 4 EMA records are sufficient to accurately estimate average pain scores; (3) binarization should be avoided if at all possible; (4) if binarization is necessary to address the goals of an analysis, statistical modelling should be performed on the original scale and should account for distributional features of the data, the results of which may be used to correctly model the binary outcome of interest, propagating relevant uncertainty in the binarization.

Conflict of interest statement

Ciprian Crainiceanu is consulting for Bayer and Johnson and Johnson on methods development for wearable and implantable technologies. The details of these contracts are disclosed through the Johns Hopkins University eDisclose system. The research presented here is not related to and was not supported by this consulting work. The remaining authors have no conflicts of interest to declare.

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Data analyzed in this article are not publicly available, though code to reproduce the simulation study is available on request to the corresponding author.

Supplemental digital content

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